Exercise 2.6

Given

$$p_x = 0.99, \qquad p_{x+1} = 0.985, \qquad {}_3p_{x+1} = 0.95, \qquad q_{x+3} = 0.02,$$

calculate

(a) p_{x+3},

(b) ${}_2p_x$,

(c) ${}_2p_{x+1}$,

(d) ${}_3p_x$,

(e) ${}_{1|2}q_x$.

Solution

(a)

Using the relationship

$$p_{x+3} = 1 - q_{x+3},$$

we get

$$\begin{aligned} p_{x+3} &= 1 - 0.02 \\ &= 0.98. \end{aligned}$$

Therefore,

$$\boxed{p_{x+3} = 0.98}.$$

(b)

The two-year survival probability from age x is

$${}_2p_x = p_x p_{x+1}.$$

Thus,

$$\begin{aligned} {}_2p_x &= 0.99(0.985) \\ &= 0.97515. \end{aligned}$$

Therefore,

$$\boxed{{}_2p_x = 0.97515}.$$

(c)

We are given

$$ {}_3p_{x+1} = 0.95. $$

Also,

$$ {}_3p_{x+1} = p_{x+1}p_{x+2}p_{x+3}. $$

Since

$$ {}_2p_{x+1} = p_{x+1}p_{x+2}, $$

we can write

$$ {}_3p_{x+1} = {}_2p_{x+1}p_{x+3}. $$

Therefore,

$$ \begin{aligned} 0.95 &= {}_2p_{x+1}(0.98), \\ {}_2p_{x+1} &= \frac{0.95}{0.98}, \\ {}_2p_{x+1} &= 0.969387755. \end{aligned} $$

Thus,

$$ \boxed{{}_2p_{x+1} = 0.969388}. $$

(d)

The three-year survival probability from age x is

$$ {}_3p_x = p_xp_{x+1}p_{x+2}. $$

Since

$$ {}_2p_{x+1} = p_{x+1}p_{x+2}, $$

we can write

$$ {}_3p_x = p_x\, {}_2p_{x+1}. $$

Using the value from part (c),

$$ \begin{aligned} {}_3p_x &= 0.99(0.969387755) \\ &= 0.959693878. \end{aligned} $$

Thus,

$$ \boxed{{}_3p_x = 0.959694}. $$

(e)

The probability ${}_{1|2}q_x$ means that the life survives one year and then dies within the following two years. Hence,

$$ {}_{1|2}q_x = p_x \, {}_2q_{x+1}. $$

Also,

$$ {}_2q_{x+1} = 1 - {}_2p_{x+1}. $$

Therefore,

$$ \begin{aligned} {}_{1|2}q_x &= p_x \left(1 - {}_2p_{x+1}\right) \\ &= 0.99(1 - 0.969387755) \\ &= 0.030306122. \end{aligned} $$

Thus,

$$ \boxed{{}_{1|2}q_x = 0.030306}. $$